



DELTA-SIGMA ASIC Development

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Team 33



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1. INTRODUCTION

- Delta-Sigma ADC falls in the category of oversampling data converters that achieve higher Effective Number of Bits (ENOBs) by sampling the signal well above the Nyquist sampling rate. A simple oversampling ADC gives us 3dB rise in the SQNR with an octave increase in sampling rate, but delta sigma ADC shapes the white quantization noise to reduce the noise in the signal band at the cost of increased out of band noise giving us increased SQNR as compared to oversampling per se.
- The objective is to achieve ENOBs greater than 16 at target Nyquist sampling frequencies between 0.5 ksp/s to 2 ksp/s, to explore and simulate various DSM architectures (DTDSM, CTDSM) and complete the ASIC design flow of the decimation filter for Delta-Sigma ADC.

2. ARCHITECTURE AND METHODOLOGY

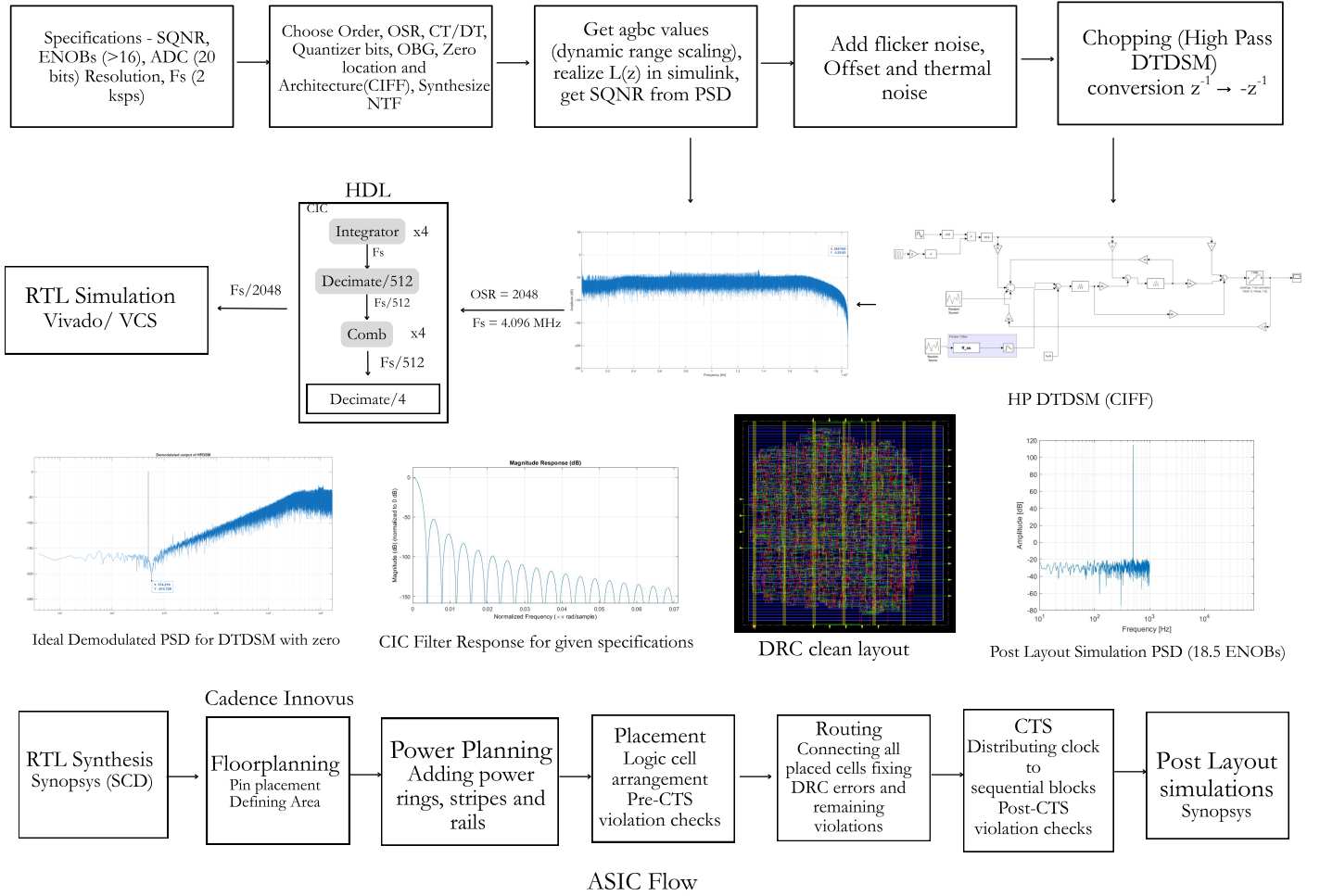


Fig. 1: Complete Flow Diagram

3. MODULATOR STUDY

A. Types Explored (DT, CT, Hybrid)

- DTDSMs and CTDSMs were explored and simulated using simulink models. Discrete DSMs sample the signal upfront before the loop and continuous DSMs sample the signal after the loop filter $L(s)$. CT is preferred for low power and higher speed. For DT, the transfer function of the S/H circuit's kT/C noise is STF, hence it degrades the SQNR. The effect of thermal noise is lesser in CTDSM since sampling is after $L(s)$. Hybrid is an optimal compromise between eliminating jitter and speed. DT at low signal frequencies give higher coefficient accuracy and well controlled integrator behaviour, whereas CT modulators at low frequency suffer from integrator finite gain effects and DAC pulse shape errors.

- CTDSM have an inherent anti-aliasing property which offers notches at F_s , thus obviating the design of an anti-alias filter required in the DTDSM before sampling. CTDSMs also suffer from Excess-Loop Delay compromising the NTF stability and doing ELD compensation is an overhead. Hybrid DSM uses CT stage at input to provide anti-aliasing benefit followed by DT stage for robust noise shaping and better component matching while eliminating clock jitter sensitivity of CT DSM.

B. ENOB vs Sampling Rate results (0.5–2 ksp/s range)

- Higher OSR translates to lower cutoff for the decimation filter, lesser quantization noise and high ENOBs at the cost of high frequency operation of sampling circuit. Given that the maximum oversampling frequency is 10MHz and the worst case Nyquist sampling frequency is 2ksp/s, the maximum calculated OSR is 5000. To ease the design of the decimation filter, we choose the OSR to be a power of 2, capping maximum OSR to 4096.
- For a fixed S/H circuit's frequency, as the sampling rate increases from 0.5ksp/s to 2ksp/s, the input bandwidth increases, hence OSR reduces, thus reducing the SQNR. All simulations ahead are done for the worst case sampling frequency of 2ksp/s.
- The expected SQNR is calculated from the theoretical formula:

$$\text{SNR} = 6.02N + 1.76 + (2L + 1)10\log(\text{OSR}) + 10\log\left(\frac{2L + 1}{\pi^{2L}}\right)$$

where N is the number of quantization bits and L is the order of the modulator and is compared with simulink model of DSM. ENOBs are obtained by $(\text{SNR}(\text{dB}) - 1.76)/6.02$. Obtained SQNR is measured from simulink models using coherent sampling ($F_{in} = kF_s/N$) and a periodic hanning window is used to avoid spectral leakage. Last NFFT samples are taken, as output stabilizes by then.

- The table below confirms the expected trend. For order 1, 2, and 3 modulators, the peak SQNR scales as M^3 , M^5 , and M^7 respectively. Therefore, an octave increase in the oversampling ratio M improves SQNR by approximately 9 dB, 15 dB, and 21 dB for first-, second-, and third-order modulators respectively. However, selecting a higher value of M increases the complexity of the decimation filter design and demands higher-frequency circuits.
- The order of the DSM was chosen to be two, as first order did not produce desired ENOB, while higher order DSMs > 2 are not unconditionally stable because they suffer from lower Maximum Stable Amplitude (MSA) and cannot be brought back once they saturate.

Order	OSR	Obtained SQNR (1-bit)	Expected SQNR (1-bit)	MSA	Obtained ENOBs
1	128	65.01	65.83	0.75	11
1	256	73.5	74.86	0.75	12
1	512	83.23	83.89	0.75	14
1	1024	92.7	92.92	0.75	16
1	2048	101.2	101.95	0.75	17
1	4096	110	110.98	0.75	18
2	128	84.85	100.25	0.65	14
2	256	99.84	115.3	0.65	17
2	512	114.5	130.35	0.65	19
2	1024	130.8	145.4	0.65	22
2	2048	147	160.45	0.65	25
3	128	103.22	133.91	0.60	17
3	256	123.96	154.98	0.60	21
3	512	148.7	176.05	0.60	25

TABLE I: SQNR and ENOBs comparison for different orders and OSR values for a DTDSM

Order 2, MSA = 0.65, Used Amplitude = 0.9×MSA, DC bias = 0.9						
OSR	SQNR (1-bit)	ENOB (1-bit)	SQNR (2-bit)	ENOB (2-bit)	SQNR (3-bit)	ENOB (3-bit)
128	84.85	14	90.8	15	97.1	16
256	99.84	17	107.9	18	113.03	19
512	114.5	19	121	20	127.84	21
1024	130.8	22	135.2	23	142.91	24
2048	146.94	25	152.22	26	157	27

TABLE II: SQNR and ENOBs values for 1-bit, 2-bit, and 3-bit quantizers for a 2nd-order modulator in a DTDSM

Sampling rate (ksps)	Obtained SQNR	OSR	ENOBs
2	130.455	1024	22
1.5	136.56	4096/3	23
1	145.8	2048	24
0.5	157.56	4096	26

TABLE III: ENOB vs Sampling Rate results (0.5–2 ksps range), $F_s = 2.048$ MHz, order = 2

- The Table II reflects an expected rise of 6dB in SQNR per bit increase of quantizer, as noise power is proportional to Δ^2 , and Δ reduces by an octave per bit increase of the quantizer. We choose the number of bits for the quantizer to be 1, because the DAC's output error sequence has a transfer function of STF, and can corrupt the output spectrum. Since a single-bit quantizer can be modelled as a gain and offset stage which can be corrected in contrast to the error sequence for multi-bit DACs.
- MSA level is found as the point where SQNR saturates or quantizer input blows up. The simulated slope of amplitude vs SQNR confirms the expected value of 10dB/dec, as 3dBFS reduction in amplitude reduces the SQNR ($\propto A^2$) by 3dB.
- We have used Richard Schreiner's ToolBox [2] to realise the a, g, b, and c parameter values for the filter loop and the noise transfer function. Higher OBG (out of band gain) represents more aggressive noise shaping, leading to instability as the quantizer input increases with increase in OBG. OBG was finalized as 1.5 (Lee's Rule for 1 bit quantizer) avoiding instability while keeping noise shaping to be moderate. In accordance with results of the table, design parameters for the modulator are finalised as 2nd order, OSR 2048, OBG 1.5, 1 bit quantizer and MSA 0.6 for our input signal which produces a desirable ENOB with a safety margin.
- Adding a zero to NTF helps to minimize the in-band quantization noise. Optimal zero position is computed by minimizing noise power integral over signal band. For a 2nd-order modulator the position is $\frac{\pi}{\sqrt{3} \cdot \text{OSR}}$ [1]. Addition of zero gives SQNR improvement of 3.5dB for a second order and 8dB for a third order modulator [1]. We use the 'g' parameter to add zeros in the NTF in the feedforward topology. We observe a HD3 of < -100dBFS (SFDR), no HD2 is observed due to the odd symmetry of the mid-rise 1-bit quantizer.

C. Excess Loop Delay in CTDSM - Analysis and Compensation

- Simulation of second order CTDSM was done using NRZ DAC (as it uses lesser bandwidth than RZ, is less sensitive to jitter, and is easier to implement) and feedforward topology with resonators. Excess loop delay (ELD) was modelled and its effect on the stability of NTF was verified in simulink and using pole-zero plots and mitigated using ELD compensation. $L(z)$ was obtained by fitting the sampled sequence at the output of $L(s)$ and finding the z -transform of that sequence.
- In order to find $L(z)$ in presence of ADC delay and Dynamic Element Matching for multi-bit DSMs (DEM) logic delay (τ), we applied an NRZ impulse and ELD is modelled using the ideal delay block of delay τ . ELD pushes the NTF poles away from origin and increases order [1]. ELD compensation pulls the poles back to



Fig. 2: Obtaining $L(z)$ with ELD in Simulink to get poles of NTF in the presence of ELD

origin ensuring stable NTF. The low frequency NTF and SQNR are almost same due to very high low frequency loop-gain. SQNR drops from 146dB to 135dB (OSR=2048, with NTF zero) in the presence of ELD, and is restored after compensation.

- The continuous-time loop-gain function needed to yield the desired NTF (no excess delay) is $L(s) = \frac{k_1}{s} + \frac{k_2}{s^2}$. With ELD, the loop-gain is $L(s) e^{-s\tau}$. Delay can be compensated by multiplying $L(s) e^{-s\tau}$ by $e^{s\tau}$. We approximate the delay term $e^{s\tau}$ using a Taylor series expansion: $e^{s\tau} = 1 + s\tau + \frac{(s\tau)^2}{2!} + \dots$. We can get the compensated coefficients by truncating after the $s\tau$ term as ω/F_s is very small within the signal bandwidth (low frequency). This makes terms involving higher powers (like $s^2\tau^2$ or $s\tau$) negligible [1]. Simulink simulations verified that as τ increases, the NTF poles move away from origin, and eventually move out of unit circle, and compensation brings them back to origin.
- In order to verify the anti-aliasing property of CTDSM, we gave an input of $0.99F_s$, and since the NTF has notches at multiples of F_s , we have an inherent anti-alias filter. The PSD plot shows that $0.99F_s$ is rejected and is aliased at $0.01F_s$.

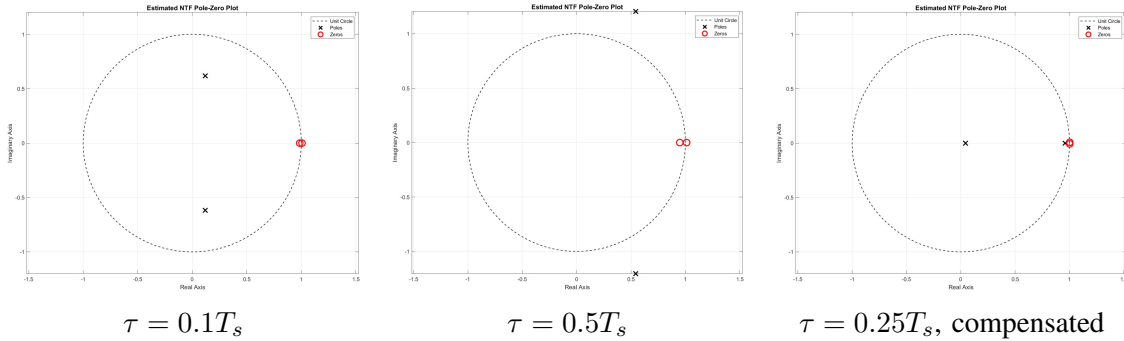


Fig. 3: NTF pole-zero plot with/without ELD Compensation

D. Selected Architecture with Justification

- In a CIFB architecture all integrator outputs have a significant input component (u) leading to higher internal swing, thus for a given UGB, the bandwidth will reduce [1], also higher swing leads to greater nonlinearity. Whereas, in CIFF, we have a fast and less precise path of a single integrator giving fast and coarse settling which is then refined by the more precise slow path of two cascaded integrators aiding settling of the loop, additionally only the last integrator output has a significant input component reducing internal signal swing for the other integrator(s) thus improving bandwidth and linearity. Hence we preferred CIFF architecture in our design. We choose CIFF DTDSM over CTDSM since we are at lower frequencies (0.25-1kps), and the the overhead of ELD compensation and clock jitter is mitigated.

4. NOISE ANALYSIS

A. Flicker Noise Impact

- To model the flicker noise, we designed a custom $1/f$ filter that uses a white-noise generator. This white noise is passed through a -10 dB/dec, $1/f$ roll-off filter (made by adding adjacent poles and zeros for an average roll-off

of -10 dB/dec) up to the flicker corner frequency, ensuring accurate $1/f$ flicker characteristics within the desired band. Beyond the cutoff frequency, a steep 200 dB/dec attenuation was applied to prevent the formation of an unrealistically high thermal noise floor. An offset voltage of $1 \mu\text{V}$ is also added.

- The effect of adding flicker noise was simulated in both CT and DT DSM where NTF contains zero. A degradation in SNR of approximately 50 – 60 dB was observed, which reduced the ENOBs below the target specification. The thermal noise floor of kT (approximately -200 dB) was modelled using an additional white-noise source with a variance of 10^{-20} . Since this thermal floor is extremely low, it does not affect the SQNR.

Comparison of CIFF DT LP DSM vs. CIFF CT LP DSM at OSR = 2048				
TYPE	SNR (dB) DT LP DSM	ENOB DT LP DSM	SNR (dB) CT LP DSM	ENOB CT LP DSM
Without flicker	144.77	24	152.22	25
With flicker	80.0	14	89.72	13

TABLE IV: Performance comparison of CIFF DT LP DSM and CIFF CT LP DSM with and without flicker noise.

B. Chopping / nested chopping / auto-zeroing comparison

- Flicker and offset corrupts the DC bins of the PSD, thus degrading SNDR. It is removed using chopping, which can be done either on system level around the complete modulator (DTDSM) or around the first integrator (CTDSM), since the other integrators' offset referred to input is ideally zero, due to infinite DC gain of the integrator. Both of these chopping techniques were simulated in simulink and verified. Op-amp model was used in simulink to simulate local chopping around an integrator in CTDSM.
- Switches used for implementation of chopping exhibit charge injection and clock feedthrough (prominent at high frequencies) at f_{chop} which is demodulated to DC creating a residual offset. Nested chopping can be used where the outer chopper (operating at lower frequency) upconverts this residual demodulated offset (of the inner chopper), and it is filtered. Injection/Feedthrough of the outer chopper is less, since it operates at a lower frequency.
- Nested chopping was simulated in Simulink, by modelling the charge injection/clock feedthrough as a pulse of frequency $F_s/2$. The outer chopper operating at $F_s/64$ modulated this residual offset and it was filtered.
- Input referred offset of the latch comparator can be reduced by using a pre-amplifier and autozeroing can be done for the pre-amplifier's offset. Auto-zeroing was simulated in simulink and results verified that the input referred offset of the comparator is reduced by A_{preamp} and offset of the preamplifier is reduced by $1/(A_{\text{preamp}} + 1)$. Sampling in autozeroing folds back the thermal noise, thus raising the noise floor and degrading SQNR [1]. The offset of the quantizer has a transfer function of NTF, so auto-zeroing isn't of much benefit specially for single-bit quantizers in contrast to Flash ADCs [1]. Auto-zeroing isn't feasible in CT integrators since they require integration in continuous time, a time-sharing ("ping-pong") technique is used to mitigate this [8]. Chopping seems to be a better-off technique to remove flicker noise and offset in both continuous and discrete time DSMs.
- In DTDSM, we use system level chopping to remove the effect of flicker noise and offset. The chopping frequency is chosen as $F_s/2$, in order to push the flicker noise and DC offset as far as possible from the signal band. Modulation at $F_s/2$ mirrors the power spectral density (PSD) about $F_s/2$. In a fully-differential version, we can just swap inputs at the clock edge using switches for modulation.
- We observed the quantization noise and flicker noise behaviour are opposite at high and low frequency regions resulting in quantization noise remaining in the signal band for LP DSM even though flicker noise is eliminated. Quantization noise shaped by the LP DSM cannot be eliminated by chopping, as it results in the peak LP DSM shaped quantization noise which is high at f_{chop} to alias back into the baseband upon demodulation, completely degrading the SNDR.
- Thus to eliminate flicker while preserving noise shaping, we modified our design to a HP DSM [4], which reverses the bowl-like NTF shape by using the transformation $z^{-1} \rightarrow -z^{-1}$ [5]. Hence, upon demodulation, only the low quantization noise region at f_{chop} aliases back into the baseband pushing higher noise shape to f_{chop}

and its harmonics that can be filtered out. The digital demodulation is done using digital logic in verilog by multiplication of the output bitstream with a sequence of $(-1)^n$. In order to realise a High Pass DSM, we used the same agbc coefficients and replaced $1/(z - 1)$ with $-1/(z + 1)$ to convert the LPDSM to HPDSM. In order to realise $-1/(z + 1)$, a mirrored integrator can be used [6].

Modulator	SNR (dB)
High-Pass DTDSM with flicker noise and system chopping around the complete modulator	137.79
CTDSM with delay and compensation, with flicker noise, offset and chopping around integrator	143.64
CTDSM with charge injection, clock feedthrough, flicker noise, offset and chopping around integrator	120.86
CTDSM with Nested Chopping with charge injection, clock feedthrough, flicker noise and offset	133.08

TABLE V: SNR Improvement with Chopping for OSR = 2048 for CTDSM, $\tau = 0.25T_s$ and DTDSM

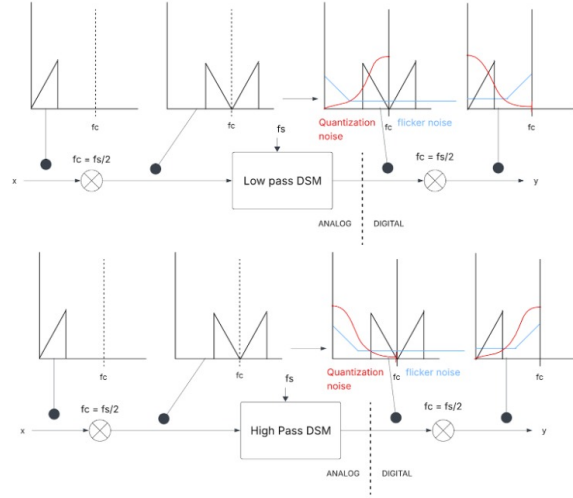


Fig. 4: Noise Shaping for High Pass and Low Pass DSM - System Level Chopping around the complete modulator

C. Final chosen technique for achieving ENOBs > 16 - $F_s/2$ modulation

- We chose the final modulator as a High pass DTDSM to mitigate flicker noise using chopping on the complete modulator, instead of chopping around an integrator. Chopping around an integrator in CTDSM is highly sensitive to parasitics. Even a small parasitic capacitance may lead to severe performance degradation due to quantization noise demodulation into the baseband [7]. The major architectural change in High Pass DTDSM [7] can be mitigated by using a mirrored integrator [6] and same agbc coefficients as previously mentioned.

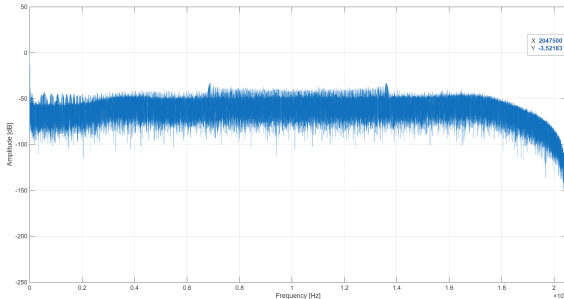


Fig. 5: HP DSM NTF shape (reflection about $F_s/2$)

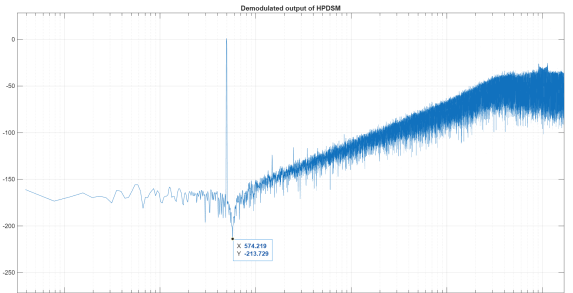


Fig. 6: Demodulated PSD for DTDSM with zero, flicker noise, offset and thermal noise

5. DIGITAL/DECIMATION FILTERS

A Delta-Sigma Modulator (DSM) produces a high-frequency, oversampled bitstream in which quantization noise is shaped out of the signal band. To obtain the final digital output at the desired sampling frequency, a decimation

filter must be applied. The decimation filter removes out-of-band noise and then reduces the sampling rate by the oversampling ratio (OSR). By choosing the decimation factor equal to the OSR, the system converts the DSM's high-rate 1-bit stream into a lower-rate multi-bit (20) digital signal while preserving in-band accuracy.

Our objective is to design a filter having 3dB cutoff of $\pi/2048$, narrow transition width and stop band attenuation of at least 80dB. The use of IIR filters was ruled out because feedback paths hinder pipelining, tighten timing constraints, and affect stability. Thus, we explored FIR and SINC filters, specifically Cascaded Integrator Comb (CIC) Filters.

A. FIR Filters

Although a single FIR filter could theoretically achieve the required response, it would be impractical to implement such a filter. It would demand a high ordered filter resulting in excessive memory usage, a large number of multipliers, higher power consumption, and increased hardware complexity.

Instead, we use a cascade of halfband FIR filters, since they achieve the desired response with much lower order, and still provide sufficient precision. Alongside they also require less number of coefficient bits to implement. Given our OSR of 2048, we implement a cascade of 11 halfband filters and perform the required downsampling. Each decimation filter should have 3dB cutoff at $\pi/2$ followed by downsampling by 2.

1) *Filter Specifications:* For our implementation, we selected a 101-tap FIR filter. This configuration provides a narrow transition band and strong stopband attenuation. Although such a filter would be challenging to implement efficiently in hardware, our goal was to evaluate its performance and verify feasibility of the design. The coefficients were provided by MATLAB Filter Designer.

Window Type	Transition Band (0–1)	Attenuation (dB)	Ripple in stopband (dB)
Blackman	0.12	-80	Between 100 and 80
Blackman-Harris	0.21	-110	Between 110 and 120
Chebyshev	0.08	-58	Between 58 and 73
Hamming	0.09	-60	Between 60 and 70
Rectangular	does not reach 40 dB	-35	—

TABLE VI: Characteristics of various window functions

Our MATLAB implementation uses the same coefficients in double precision floating point format. The results we obtained from testing different windowing functions have been logged in Table VI. We needed to select a window for a small transition band and stop band attenuation of atleast 80dB [9]. We found Chebyshev window function to satisfy the requirements.

2) *Quantization and Truncation:* The coefficients were quantized to 32-bit fixed-point (1 sign bit and 31 fractional bits) by MATLAB Filter Designer. We evaluated the 32-bit final output width performance before reducing the bit width to 20 bits.

Output growth must be controlled in cascaded FIR filters and in fixed-point we need to keep track of the decimal point as well. We first find the maximum possible sum of all the coefficients to get the maximum possible output of the first filter. This gives an estimate on output bit width of accumulator. The multiplication of M and N number of fraction bits gives us $M + N$ fraction bits and summation does not affect the number of fraction bits. We have designed the accumulator width such that we can simply take its 32 MSB bits each time. Fig. 7 shows how we perform truncation using 2 cascaded FIR filters as an example. We continue applying the same technique for the other 9 filters.

(S,N,F) : S: Number of sign bits; N: Number of integer bits; F: Number of Fraction bits

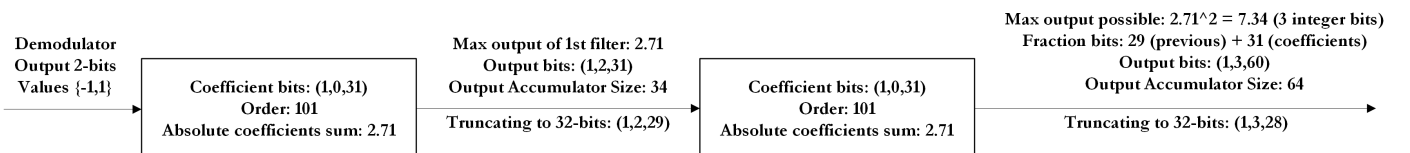


Fig. 7: Bit growth and accumulator sizing when cascading two 101-tap halfband FIR filters.

3) *MATLAB vs Verilog Simulation Results:* As observed in Table VII, MATLAB generally gives higher SNDR as compared to verilog simulations due to increase in precision. The verilog model uses 32-bit values while MATLAB is set to double precision. Truncation is done after each stage in verilog for the desired output bit width. This results in loss of information, especially at higher frequencies where the information is mainly stored in the lower bits. To tackle this problem we would need higher order filters, larger accumulator, and larger output bit width which increases hardware complexity.

Input Frequency (Hz)	SNDR, MATLAB Simulation (dB)	SNDR, Verilog Simulation (dB)
10	66.78	69.18
500	62.35	61.01
990	62.17	57.92

TABLE VII: Simulation results comparison of SNDR of FIR filter

Although further refinement of the fixed-point implementation might be possible, our MATLAB results show that even double-precision floating-point implementation did not meet the required SNDR specifications. This observation motivated us to shift from FIR filters to CIC filters, which offer lower hardware complexity and less sensitivity to bit-width limitations.

B. CIC Filter

CIC filters are hardware efficient due to their multiplier-free structures which makes them suitable for high speed filtering. CIC filters typically require droop compensation; however, when implemented in verilog, the compensation stage significantly degrades the output quality. Since the design specifications were being satisfied for certain CIC filter parameters, as confirmed through MATLAB analysis, the compensation stage was omitted. The CIC filter was found to have a linear phase in the passband, verified manually by taking impulse response and plotting the phase response in MATLAB

1) *Filter Parameters:* Below are the different combinations of the CIC filter that were tested to match our requirements. We explored cascaded CIC filters and single CIC filter configurations. N denotes the number of stages and R denotes the decimation factor of the CIC filter.

- Cascaded CIC: Table VIII presents the MATLAB and verilog performance comparison. The most satisfactory result was obtained using the configuration of two cascaded CIC filters having $N_1 = 2$, $R_1 = 512$, $N_2 = 1$, $R_2 = 4$ achieving an SNDR of 103dB which translates to 16.81 ENOBs, having a passband droop of 3.8dB. The passband droop has been calculated as the difference of amplitude in PSD of 10Hz and 990Hz input signals in dB. Peak SNDR has been calculated for an input signal frequency of 10 Hz. We have kept the differential delay M to be 1 because of power and memory minimization.

N_1	N_2	R_1	R_2	Peak SNDR (dB)	Passband Droop (dB)	ENOBs
2	2	64	32	114	8.06	18.65
3	1	128	16	93	4.33	15.15
1	2	16	128	73	4.23	11.83
3	1	16	128	76	6.23	12.33
2	1	256	8	94	3.63	15.32
2	1	512	4	103	3.82	16.81

TABLE VIII: Configuration and performance characteristics of different cascaded CIC filters

- Single CIC: We tested a variety of different single CIC filters and chose a 4th order, 512 decimation CIC as our final configuration. This provided an approximate ENOBs of 18.38. We optimized CIC filter for low passband droop, but higher ENOBs can be achieved by compromising on the passband droop as evident from Table IX.

N	R	Downsampling rate	Peak SNDR (dB)	Passband droop (dB)	ENOBs
2	2048	1	113.96	7.67	18.64
3	2048	1	117.26	11.51	19.19
2	1024	2	102.28	1.79	16.70
3	1024	2	117.12	2.86	19.16
2	512	4	90.52	0.44	14.75
3	512	4	110.47	0.66	18.06
4	512	4	112.41	0.88	18.38
4	256	8	99.28	0.22	16.20
8	256	8	106.09	0.44	17.33
16	256	8	111.36	0.88	18.21

TABLE IX: Configuration and performance characteristics of different single stage CIC filters

2) *Quantization and Truncation*: In 2 cascaded CIC configuration, we observed that any truncation results in a poor SNDR, but this was not the case for a single CIC filter. We tested three different truncation techniques for reducing our output from 38-bits to the required ADC resolution of 20-bits:

- **Truncation**: During our testing we noticed the 3 MSB bits were being occupied by the sign bits regardless of input frequency. Therefore we decided to omit 2 MSB bits and only take bits [35:16] from our output of [37:0]. All the other techniques work on truncating [35:0] to [35:16].
- **Rounding**: The output word [35:16] is rounded according to the magnitude of the discarded bits [15:0].
- **Dithering** [11]: Before truncation we add random noise generated using the difference of output given by 2 LFSRs [10] with different seeds, add it to the LSB bits, round off, and then truncate the final result from [35:0] to [35:16].

Table X lists the SNDR of the filter output at various input frequencies. For comparison, we also include the SNDR values obtained when the omitted sign bits are retained, to illustrate the impact of preserving those bits.

Truncation Method	SNDR - 10 Hz (dB)	SNDR - 500 Hz (dB)	SNDR - 990 Hz (dB)	Passband Droop (dB)
Dithering	110.13	110.13	108.88	0.88
Truncation[35:16]	112.41	112.40	111.15	0.88
Truncation[36:17]	108.52	117.80	109.73	0.88
Truncation[37:18]	105.07	120.16	104.40	0.88
Rounding	108.97	109.56	108.75	0.90

TABLE X: SNDR and droop comparison for various output quantization methods

3) *MATLAB vs Verilog Simulation Results*: Table XIV compares the MATLAB and Verilog simulation outcomes. MATLAB yields higher SNDR because it operates in double-precision arithmetic, which significantly reduces truncation effects.

Frequencies (Hz)	SNDR, MATLAB Simulation (dB)	SNDR, Verilog Simulation (dB)
10	115.05	112.41
500	115.85	112.40
990	113.06	111.15

TABLE XI: Simulation results comparison of SNDR

6. HDL DESIGN

A. RTL architecture, block descriptions

- **Demodulator**: This block converts the incoming 1-bit bitstream into a 2-bit signed output using a two-state FSM. In state 1, the input is appended with a 0, effectively multiplying it by +1. In state 2, a 0 input produces 00, while a 1 input produces 11 (−1 in signed form), corresponding to multiplication by −1. Thus, the bitstream is

multiplied by a $F_s/2$ periodic square-wave sequence $(-1)^n$. The demodulator runs at the sampling frequency F_s , producing outputs at the same rate.

- **Integrator:** The integrator stage forms the first section of the CIC filter and implements a discrete-time accumulator. It is a register-based design where an OW-bit register stores the running sum of the sign-extended input. The output is simply the accumulator's updated contents.
- **All_Integrators:** The All Integrators module forms the CIC integrator stage by cascading N basic integrators to produce an N -th order running sum. Each stage feeds the next, with enable signals chained through internal `out_ready` signals, while the first stage uses the external enable input. The final stage output provides the overall accumulated result.
- **Downsampler:** This block performs sample-rate reduction using a mod- R counter and a data-hold register. Each time the counter reaches R , the current `input_data` is captured into the output register, the counter resets, and `out_ready` pulses high for one cycle to indicate a new decimated sample and to enable the next comb stage.
- **Comb:** This block implements the CIC comb stage, computing the discrete-time difference between the current input and the previous sample. A single register stores the one-cycle delay, and the output holds the subtraction result.
- **All_Combs:** This block instantiates a cascaded chain of N first-order comb stages. Each stage feeds the next, with the first stage driven by the downsampler output. The downsampler asserts the initial enable signal, and each stage's `out_ready` becomes the enable for the following stage.
- **Cic_Top:** This module integrates the `all_integrator`, `decimator`, and `all_comb` blocks to form an N th-order CIC decimator. The integrators compute running sums, whose outputs are downsampled by a factor R . The decimated stream then passes through N comb stages. Valid and ready signals between stages (`integrator_ready`, `downsampler_ready`, `out_ready`) ensure correct data flow across the rate-change boundary.
- **Full_Filter:** This module implements a two-stage decimation cascade. The first stage is a full CIC filter (`cic_top`) that performs filtering and decimation by R_1 , producing `cic_out` and `out_ready_cic` at a rate of F_s/R_1 . Next, a simple downsampler that applies decimation by R_2 . The final output `dout` is thus generated at $F_s/(R_1R_2)$.
- **Top:** This top-level module processes the 1-bit oversampled output of a sigma-delta ADC. The demodulator first converts the single-bit stream into a 2-bit signed value, which is then passed to a two-stage decimation filter comprising a CIC decimator followed by a downsampler. Finally, the decimated output is truncated to a user-specified bit width.

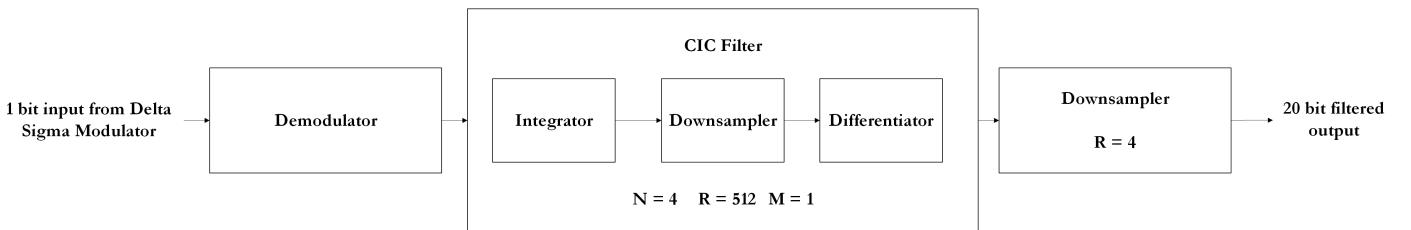


Fig. 8: HDL Code Block Diagram

B. Simulation results and observations

1) *Simulation Setup:* The testbench processes a 1-bit input stream representing the output of the simulated delta sigma modulator containing a 500Hz signal. The input file fed to the test contains over 4 million binary samples. The system operates at a 4.096 MHz clock frequency and implements substantial signal reduction through a 4-stage CIC filter with a decimation factor of 512, followed by an additional 4× downsampler, which resulted in a total decimation factor of 2048. This configuration transforms the high-rate modulated input into a 20-bit signed output stream at 2 kSPS.

2) *Simulation Results:* The simulation produced 2000 output samples which were analyzed in MATLAB. The psd of the output showed a peak at 500Hz and recorded an SNDR of over 112, which translate to over 18.4 ENOBs.

C. Timing analysis results

The static timing analysis report, generated with a 250 ns clock period (4 MHz target frequency), reveals exceptional timing performance with substantial margin. The critical path originates at `demodulator_inst/ps_reg` and terminates at `full_filter_inst/cic_inst/integrators/integrator_gen[0].integrator/acc_reg[37]`, constituting a 38-bit ripple-carry propagation through the CIC filter's primary integrator adder. The data arrival time measures 3.39 ns, comprising 0.35 ns of initial combinational logic and 3.04 ns of adder chain propagation through 31 full-adder cells with an average delay of 0.098 ns per cell. With a required time of 249.99 ns (accounting for -0.01 ns library setup time), this yields a positive slack of 246.60 ns, representing a utilization of only 1.36 percent of the available clock period. The four subsequent reported paths (to bits 36-33) exhibit similar characteristics with slacks ranging from 246.64 ns to 246.87 ns, confirming timing closure with over 98.6 percent margin across all critical paths. (Note: This analysis was done with UMC65 nm library)

7. ASIC FLOW

A. Synthesis Results

In Synthesis Stage our RTL code is converted into a gate-level netlist using technology libraries (.lib) and physical layout files (.lef) from the PDK (we have used UMC's 65nm technology) in Synopsys Design Compiler (SDC). The TCL and SDC files were customized, timing constraints, relative paths were applied, and it was observed how synthesis optimizes area, power, and timing.

Instance	Cell Count	Total Area (μm^2)	Clock (ns)	Critical Path Delay (ns)	Slack (ns)
top	1509	8070.12	250	3.12	246.87

TABLE XII: Area and Timing Results

B. Layout Summary

Layout was performed in Cadence Innovus by following the major stages: Floorplanning, Power Planning, Placement, CTS, and Routing. We began by creating the floorplan by specifying the core area, placing the input and output pins, and designing the power network (power grids, rings, and stripes) to ensure stable power delivery across the chip. Next, we performed placement of the standard cells, followed by Clock Tree Synthesis (CTS) to distribute the clock to sequential elements while meeting timing constraints. After CTS, the signals were routed while ensuring timing closure. Timing violations are addressed through optimization steps performed during the pre-CTS and post-CTS stages.

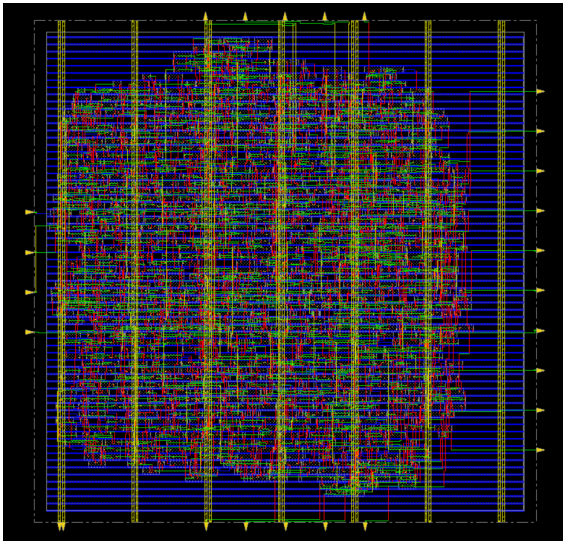


Fig. 9: DRC clean Layout

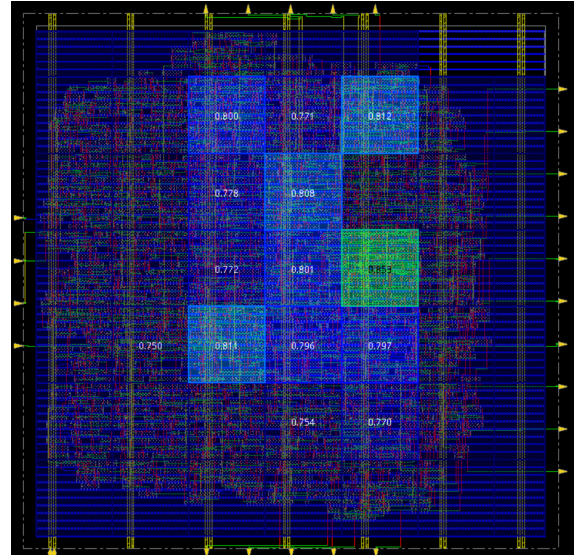


Fig. 10: Density Map

Core Utilization (%)	Average Module Density	Pin Density
53.03	0.534	0.1486

TABLE XIII: Utilization Report

C. Post-Layout Simulation Results

The final SNDR and ENOBs calculation was done using the output netlist and .sdf file generated by Cadence Innovus (Post-Layout). The simulated SNDR and ENOBs are given below. The passband droop is $1.1dB$. The obtained ENOBs meet the specification of > 16 ENOBs with a 20-bit resolution ADC, and sampling rate of $4.096MHz$.

Frequencies (Hz)	SNDR, Post-Layout Simulation (dB)	ENOBs
10	112.4	18.4
500	113.64	18.6
990	111.05	18.15

TABLE XIV: Post-Layout Simulation results

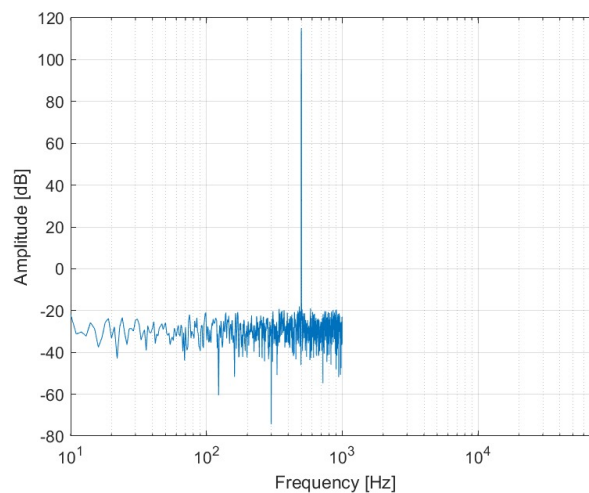


Fig. 11: Post-Layout Simulation PSD

8. CHALLENGES FACED

- HPDSM cannot be simulated at low input frequencies because the STF is high-pass, so noise shaping is not observed properly. Input should be given either after $F_s/2$ modulation or by adding $F_s/2$ in the input frequency.
- Simulating pole movement of NTF in the presence of ELD τ , which was done by injecting a delayed impulse NRZ pulse $p(t - \tau)$ through $L(s)$, sampling it and then least square fitting the output sequence, and finding its z-transform to evaluate $L(z)$.
- For visualisation of output waveform after filtering, sampling frequency, $F_s = 2kHz$ should be a multiple of the input frequency so that no false interpretations are made when MATLAB interpolates peaks.
- Several challenges arose during the design process, including quantizing FIR tap coefficients, deciding how to truncate the filter output, and managing bit growth when cascading multiple halfband filters. Experimentally determining the impulse response of the LTV CIC filter also added complexity. Many CIC configurations were evaluated, varying both decimation factors and the number of stages, while balancing the trade-off between passband droop and achievable SNDR to meet the required specifications.

9. NEW LEARNINGS

- Adding 'g' parameter in CIFF can also realise zero along with the traditional CRFF topology. Coherent sampling frequency should be very precisely specified in fraction in simulink to avoid smearing.
- In order to decouple flicker variance and thermal kT noise floor of -200dB, flicker was made 0 after the corner frequency and thermal noise was added separately to have degree of freedom on variance of both, thermal noise floor and flicker.
- In order to convert LPDSM to HPDSM, only the integrator needs to be mirrored, agbc coefficients remain same. Periodic hanning window needs to be used instead of normal hanning window for no spectral leakage. In the presence of very high thermal floor, effect of zero is removed, also noise shaping at lower frequencies is lost, and we get the usual 3dB rise in SQNR with octave rise in OSR.
- A gain block of F_s needs to be placed before the $1/s$ block in CTDSM simulink model since the DAC pulse shape has area in terms of T_s . While modulating the input ($F_s/2$ modulation), it should be sampled first and then multiplied with square wave to avoid synchronization issues in simulink, which is in line with the practical implementation of sampling and then swapping inputs at the clock edge in a fully-differential version.
- Digital demodulation can be done in verilog by pointwise multiplication with $F_s/2$ sequence of $(-1)^n$ to avoid synchronization issues. Multibit DACs can be modelled by a gain stage of $(FS/(2^N - 1))$ after a n-bit quantizer.
- CIC filter is a linear time-variant system, thus its impulse response cannot be constructed directly by exciting the filter with $\delta[n]$. For an N decimation filter, we need to excite the system with $\delta[n], \dots, \delta[n - N + 1]$ independently, and thus obtain all the required taps to construct its impulse response. For an R -stage, N -decimation CIC filter, we obtain $NR - 1$ taps. For a 3-stage, 32-decimation CIC filter, we will obtain $h[0], h[32], h[64], \dots$ for $\delta[n]$, $h[-1], h[31], h[63], \dots$ for $\delta[n - 1]$, and so on. After obtaining all these taps, we can stitch them together and obtain the required impulse response $h[n]$ for the LTV system, and analyze its magnitude and frequency response. The frequency response was found from HDL code using this method and it resembled the MATLAB frequency response. The impulse response of $SINC^3$ ($N = 32$) and $SINC^2$ ($N = 1024$) were verified from HDL.

10. LIMITATIONS AND RISKS

- In the presence of very high thermal floor, use of zero may not be beneficial. Using CTDSM may pose risk of instability of NTF if the compensated coefficients vary. DTDSM implementation may require a high value of the sampling capacitor.
- FIR filters are limited by coefficient quantization, since long decimal coefficients must be rounded to fit within a fixed bit width, affecting passband accuracy and cutoff placement. Narrow accumulators can overflow for large inputs, producing distortion. Additionally, the truncation or rounding that occurs at every multiplication introduces quantization noise that accumulates across long or cascaded FIR stages.
- In our final CIC design, we safely truncate the output from bits $[37 : 0]$ to $[35 : 16]$ because the demodulator provides only ± 1 inputs, ensuring that the CIC integrators never reach their theoretical worst-case growth. However, if digital chopping (demodulation) is not used, such aggressive truncation should be avoided, as the CIC may accumulate much larger values and potentially overflow.
- PVT variations may alter delays in the filter introducing additional timing jitter. This jitter would corrupt the output spectrum and degrade the SNDR. SS (Low Vdd, High T) exhibits the highest degradation.

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